



BME513 - Inteligencia Artificial en Salud

Fuzzy Logic and ANFIS

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Learning goals

- Explain the basic idea of fuzzy sets and fuzzy rules.
- Distinguish Mamdani and Takagi-Sugeno-Kang inference.
- Interpret a TSK model as a soft partition with local models.
- Describe ANFIS as a trainable neuro-fuzzy architecture.
- Understand the hybrid learning idea: least squares + gradient descent.

Fuzzy logic

Classical logic uses hard truth values:

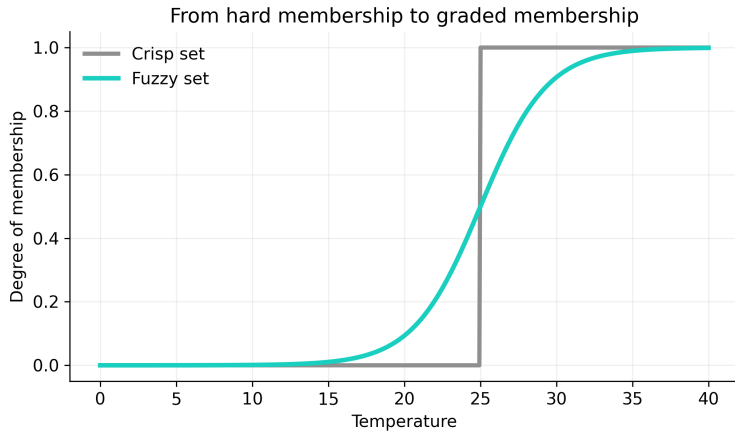
$$\text{true} = 1, \quad \text{false} = 0.$$

Fuzzy logic allows graded truth:

$$\mu_A(x) \in [0, 1].$$

This is useful when concepts are gradual: young, warm, high risk, similar.

Crisp vs fuzzy membership



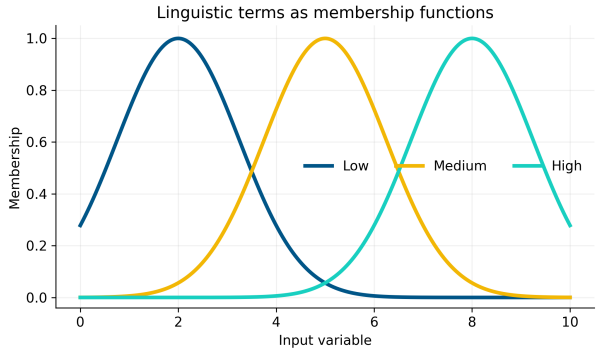
Linguistic variables

A linguistic variable is described by terms:

Temperature $\in \{\text{low, medium, high}\}$.

Each term is represented by a membership function.

Rules operate on these linguistic terms.



Fuzzy rule base

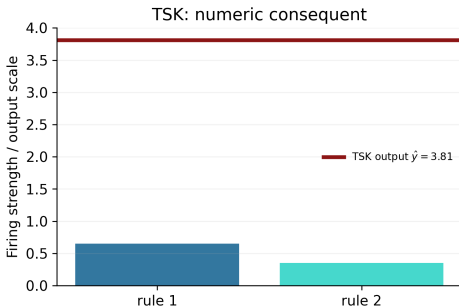
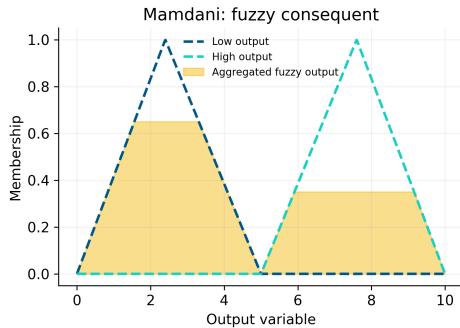
A fuzzy system is built from rules:

$$\mathcal{R}_i : \quad \text{IF } x_1 \text{ is } A_1^{(i)} \text{ AND } \dots \text{ AND } x_d \text{ is } A_d^{(i)} \text{ THEN } \dots$$

The antecedent computes a firing strength:

$$w_i = T \left(\mu_1^{(i)}(x_1), \dots, \mu_d^{(i)}(x_d) \right) .$$

Mamdani vs TSK



Mamdani inference

Mamdani rules have fuzzy consequents:

$$\mathcal{R}_i : \text{IF } x \text{ is } A_i \text{ THEN } y \text{ is } B_i.$$

Typical pipeline:

1. Fuzzify inputs.
2. Compute rule firing strengths.
3. Apply implication to output fuzzy sets.
4. Aggregate rule outputs.
5. Defuzzify.

$$\hat{y} = \frac{\int_Y y \mu_B(y) dy}{\int_Y \mu_B(y) dy}.$$

TSK fuzzy model

Takagi-Sugeno-Kang rules have numeric consequents:

$$\mathcal{R}_i : \text{ IF } x_1 \text{ is } A_1^{(i)} \text{ AND } \dots \text{ AND } x_d \text{ is } A_d^{(i)} \\ \text{ THEN } f_i(\mathbf{x}) = \theta_0^{(i)} + \theta_1^{(i)} x_1 + \dots + \theta_d^{(i)} x_d.$$

The antecedent divides the input space; the consequent models each region.

TSK prediction

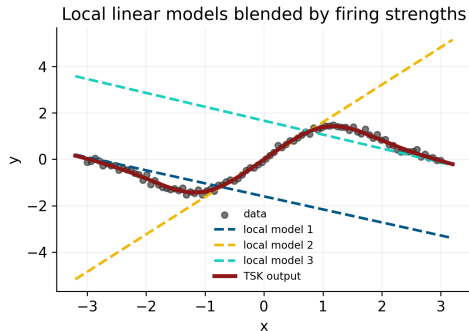
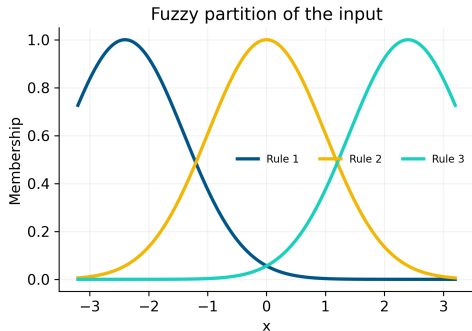
For an input $\mathbf{x}$:

$$w_i = \prod_{j=1}^d \mu_j^{(i)}(x_j), \quad \bar{w}_i = \frac{w_i}{\sum_{r=1}^n w_r}.$$

The final prediction is a normalized weighted average:

$$\hat{y} = \sum_{i=1}^n \bar{w}_i f_i(\mathbf{x}).$$

TSK as local modeling



What must be learned?

A TSK model has three design questions:

- **Premise parameters**: where are the membership functions and how wide are they?
- **Consequent parameters**: what linear model belongs to each rule?
- **Structure**: how many rules and which inputs should be used?

When these elements are learned from data, we often call the model **neuro-fuzzy**.

ANFIS

ANFIS: Adaptive Network-based Fuzzy Inference System.

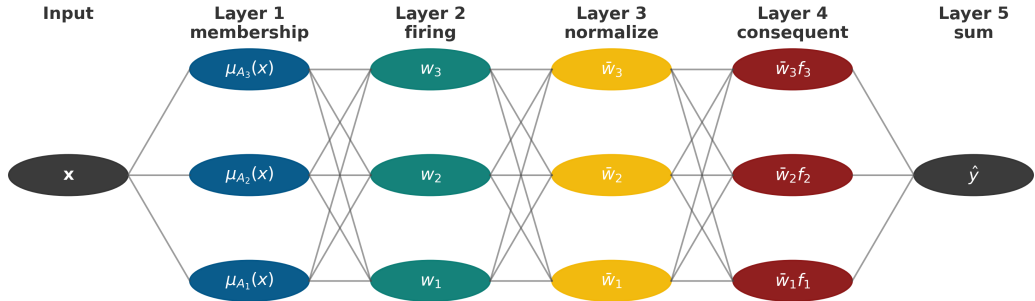
It represents a TSK fuzzy system as a layered adaptive network.

Forward pass: fuzzy inference.

Backward/update step: parameter learning.

ANFIS architecture

ANFIS as a trainable TSK fuzzy system



Layer view

Layer	Computation	Parameters
1. Membership	$\mu_j^{(i)}(x_j)$	premise: centers, widths
2. Firing	$w_i = \prod_j \mu_j^{(i)}(x_j)$	none
3. Normalize	$\bar{w}_i = w_i / \sum_r w_r$	none
4. Consequent	$\bar{w}_i f_i(\mathbf{x})$	consequent coefficients
5. Output	$\hat{y} = \sum_i \bar{w}_i f_i$	none

Hybrid learning

ANFIS commonly uses two coupled updates:

1. Fix premise parameters and estimate consequents by least squares.
2. Fix/update consequents and adjust premise parameters by gradient descent.

Why? The model is linear in consequents, but nonlinear in premise parameters.

Consequent learning

For fixed normalized firing strengths:

$$\hat{y}_k = \sum_{i=1}^n \bar{w}_i(\mathbf{x}_k) \left(\theta_0^{(i)} + \theta_1^{(i)} x_{k1} + \cdots + \theta_d^{(i)} x_{kd} \right).$$

For each observation k , one row of the design matrix is:

$$\phi_k = [\bar{w}_{1k} \quad \bar{w}_{1k} x_{k1} \quad \cdots \quad \bar{w}_{1k} x_{kd} \quad \bar{w}_{2k} \quad \bar{w}_{2k} x_{k1} \quad \cdots \quad \bar{w}_{nk} x_{kd}].$$

This can be written as a linear regression:

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_N \end{bmatrix} = \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_N \end{bmatrix} \begin{bmatrix} \theta_0^{(1)} \\ \theta_1^{(1)} \\ \vdots \\ \theta_d^{(n)} \end{bmatrix}, \quad \hat{\mathbf{y}} = \Phi \boldsymbol{\theta}, \quad \hat{\boldsymbol{\theta}} = (\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{y}.$$

Premise learning

Premise parameters control the fuzzy partition:

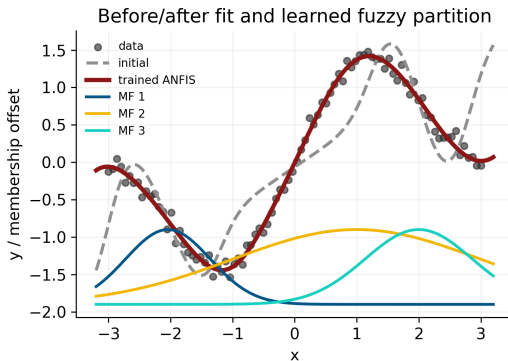
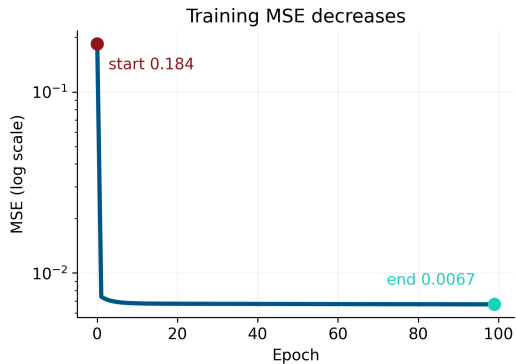
$$\mu_i(x) = \exp \left[-\frac{1}{2} \left(\frac{x - c_i}{\sigma_i} \right)^2 \right].$$

They are updated to reduce:

$$E = \frac{1}{N} \sum_{k=1}^N (y_k - \hat{y}_k)^2.$$

Moving centers and widths changes which local model is trusted.

Training example



ANFIS strengths and caveats

Strengths	Caveats
Interpretable fuzzy rules	rule explosion in high dimension
Local nonlinear modeling	sensitive to initialization
Hybrid learning is efficient	can overfit with many rules
Expert knowledge can be injected	structure selection is nontrivial

ANFIS is most attractive when accuracy and interpretability both are needed.

See [06-anfis-example.ipynb](#)

Summary

- Fuzzy sets replace hard membership by graded membership.
- Fuzzy rules combine linguistic antecedents with consequents.
- Mamdani is highly interpretable but needs defuzzification.
- TSK uses numeric local models and weighted averaging.
- ANFIS trains a TSK system with a neural-network-like architecture.
- Hybrid learning exploits linear consequents and nonlinear premises.